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Special Collection: Disordered Materials at the Atomic Scale

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ABSTRACT

Entropy stabilized oxide of $MgNiCoCuZnO_5$, also known as J14, is a material of active research interest due to a high degree of lattice distortion and tunability. Lattice distortion in J14 plays a crucial role in understanding the elastic constants and lattice thermal conductivity within the single-phase crystal. In this work, a neuroevolution machine learning potential (NEP) is developed for J14, and its accuracy has been compared to density functional theory calculations. The training errors for energy, force, and virial are 5.60 meV/atom, 97.90 meV/Å, and 45.67 meV/atom, respectively. Employing NEP potential, lattice distortion, and elastic constants is studied up to 900 K. In agreement with experimental findings, this study shows that the average lattice distortion of oxygen atoms is relatively higher than that of all transition metals in entropy-stabilized oxide. The observed distortion saturation in the J14 arises from the competing effects of minimum site distortion, which increases with increasing temperature due to enhanced thermal vibrations, and maximum site distortion, which decreases with increasing temperature. Furthermore, a series of molecular dynamics simulations up to 900 K are performed to study the stress–strain behavior. The elastic constants, bulk modulus, and ultimate tensile strength obtained from these simulations indicate a linear decrease in these properties with temperature, as J14 becomes softer owing to thermal effects. Finally, to gain some insight into thermal transport in these materials, with the so-developed NEP potential, and using non-equilibrium molecular dynamics simulations, we study the lattice thermal conductivity (κ) of the ternary compound $MgNiO_2$ as a function of temperature. It is found that κ decreases from 4.25 $W\,m^{-1}\,K^{-1}$ at room temperature to 3.5 $W\,m^{-1}\,K^{-1}$ at 900 K. This suppression is attributed to the stronger scattering of low-frequency modes at higher temperatures.

I. INTRODUCTION

The discovery of high entropy alloys (HEAs) has ushered in a new paradigm in materials design, enabling fine-tuning of configurational entropy and the prospect of influencing the phase stability of solid solutions.[1,2] This breakthrough has opened avenues for tailoring material properties through entropic engineering, thereby expanding the compositional landscape accessible to alloy design. These high entropy alloys intrinsically exhibit lattice distortion effects and offer a vast compositional space, presenting opportunities for tailoring their thermal, mechanical, and catalytic properties through judicious compositional tuning. The synergistic coupling between the configurational entropy and the resultant lattice distortions enables the engineering of desirable property profiles, thereby expanding the functionality envelope accessible within this novel class of materials. Entropy-stabilized oxides represent a subset of high entropy materials, comprising an oxygen anion sublattice and five or more metallic cations, typically present in equiatomic or near-equiatomic proportions. These multi-principal component oxides exhibit a high degree of configurational disorder on the cation sublattice, facilitated by the maximization of configurational entropy. Rost *et al.* successfully synthesized a single-phase high entropy oxide crystallizing in the rock salt structure by incorporating five equiatomic metallic species.[3] Bérardan *et al.* reported that the high entropy oxide system and its compositional derivatives exhibit remarkable functional properties, including colossal dielectric constants and superionic conductivity.[4] Since then, several experimental techniques have been developed for synthesizing, fabricating, and characterizing single-phase high entropy oxide.[5–8] Despite various methods of synthesis and fabrication techniques, probing multi-elemental composition using x-ray and electron microscopy still remains a hindrance to fully resolve the local atomic arrangements, distortions, and compositional variations in entropy-stabilized oxide.[9] The lattice distortion is inevitable in high entropy and entropy-stabilized alloys, and several works are being done to quantify and characterize it.[10–17] One of the primary difficulties in measuring lattice distortion experimentally is that these techniques provide ensemble-averaged information. For instance, using techniques such as x-ray diffraction, the local lattice strain due to off-site atomic displacements is determined by the Debye–Waller factor; considering high entropy alloys where there is a disordered electronic environment, the atomic form factor is very challenging to measure, which in turn leads to difficulty in measuring local lattice strain.[10,18,19] Another advanced technique to measure the local static distortion in high entropy alloys uses diffuse scattering, but the local lattice distortions are not immediately accessible without extensive modeling.[10] Song *et al.* demonstrated that the atomic size mismatch evaluated with the empirical atomic radii is not accurate enough to describe the local lattice distortion for refractory high entropy alloys, particularly for NbTiV alloy the lattice distortion of 0.085 is obtained using empirical atomic radii but the lattice distortion of 0.18 is obtained using ab initio molecular dynamics.[12] Due to the large compositional and configurational sampling space, computational cost on the size of single-phase high entropy oxide limits the density functional theory calculations. Using classical simulations, Anand *et al.* recently quantified lattice distortion in J14 and concluded that configurational entropy plays a crucial role in stabilizing the ceramic phases and lowering the free energy of mixing.[11] This study required many configurations to shed light on the structural behavior and stability of J14. Additionally, experimental techniques have limitations in resolving the contributions of different atomic species to the overall distortion, making it difficult to decouple the individual effects. Furthermore, ambient temperature and pressure conditions may limit understanding of how lattice distortion evolves under specific conditions for tailoring the material properties. A high-pressure study of J14 by Cheng *et al.* has shown that symmetry-breaking phase transition occurs only after ∼43 GPa due to bond bending and distortion.[19] Similarly, Rost *et al.* investigated the thermal and mechanical properties of J14 and found mechanical softening of J14.[20] The thermal conductivity remained almost constant at around 2.5 W m^{-1} K^{-1} across various temperatures. In light of this, computational and modeling work is, in principle, beneficial for such study. State-of-the-art numerical modeling techniques such as cluster expansion method, virtual crystal approximation or coherent potential approximation, and the special quasi-random structures method are being used to study the local disordered structures for high entropy alloys but are far from the experimental observation.[21–24] The cluster expansion method has limitations because it truncates the total energy of the system, assumes rigid lattice constants, and challenges modeling finite temperature effects.[25] Though virtual crystal approximation has great computational speed in modeling high entropy alloys, it generally lacks calculating mixing enthalpy, the volume of the solid solution, and other thermal properties like thermal conductivity.[26,27] With the advent of machine learning, works related to high throughput screening of structure-property relationships of high entropy alloys are on the rise, although such studies are only targeted for some specific properties. Similarly, maximum entropy distribution has also been used to model the high entropy alloys and study distortion based on the optimized configurations; the optimization procedure requires interatomic potential.[24] It is still a challenge to have a clear physical picture at the atomistic level owing to a lack of interatomic potential to describe statistical thermodynamic properties at high temperatures. Having the interatomic potential to describe the multi-component system is beneficial in the sense that it provides a bridge between the accuracy that the density functional theory (DFT) provides and the speed that molecular dynamics confer.[28]

The classical interatomic potential is based on the potential function and relies on a physical understanding of the interatomic bonding in the material.[29] Such potential models are highly approximate and contain a few adjustable parameters, thus limiting their prediction accuracy, especially for systems with both metallic and covalent bonding.[29,30] In contrast to the classical interatomic potential, machine learning interatomic potential approximates the potential energy surface to represent the locality of atomic interactions and the invariance of energy under translation and rotation.[29] Due to the random solid solution of cations on the high entropy oxide, there are many local atomic environments for the cations and anions. Therefore, it is almost impossible to explore all configurational space. Because machine learning potential is interpolable for a wide range of compositions, only a selected number of density functional theory calculations based on different compositions is sufficient for training machine learning potential.[31] Many types of machine learning interatomic potential forms are present in the literature. The central idea is to decompose total energy into contributions from each atom based on local environments, and this is usually composed of two parts: the functional form called the descriptors that describe the local environment and the regressor that maps the local environment to the energy and its derivative.[32,33] For instance, Behler introduced a neural network as a regressor to map the local environment to energy. Similarly, Bartok *et al.* used Gaussian process regression to build a machine learning potential, the Gaussian Approximation Potential (GAP).[34] Several other machine learning potentials, such as spectral neighbor potential (SNAP) and moment tensor potential (MTP), use linear regression as a regressor.[35,36] Other machine-learned interatomic potentials utilize deep neural networks and graph neural networks as regressors; however, the central idea is the same.[37,38]

Recently, using a separable natural evolution strategy, Fan *et al.* developed an evolutionary neural network potential (NEP).[39,40] NEP employs the descriptor by Behler symmetry function and uses the smooth overlap of atomic potentials (SOAP).[41,42] With the implementation of graphical processing units (GPU) for the training of NEP and the molecular dynamics (MD) simulation, the efficiency and accuracy of NEP have been demonstrated by comparing to several other machine learning (ML) potentials, including DeePMD, GAP, and MTP.[34,36,37,39] The NEP model has been successful in describing the properties of various materials, including superconductivity and superionic behavior of Li–Al compounds under high pressure, mechanical properties and thermal conductivities of two-dimensional network structures of C60 molecules, and quantum corrections of thermal conductivities of amorphous Si.[43–48] In general, NEP is a great framework, as implemented in the graphical processing units of molecular dynamics (GPUMD), which provides an accurate and efficient tool for investigating the properties of materials.

The objective of developing the neuroevolution potential is thus to study the static and dynamic lattice distortion, evaluating the elastic constants using stress–strain response and the lattice thermal conductivity of alloys. This paper is organized as follows: the first section includes the NEP model, construction of datasets, NEP training, and hyperparameter setting. In the second section, the training accuracy is evaluated, followed by results on lattice distortions and stress–strain as a function of temperature. Finally, the discussion of results is followed by the conclusion.

II. NEUROEVOLUTION POTENTIAL MODEL

A. NEP model

The potential energy of the NEP model is expressed as a summation of site energy based on the set of descriptor components $\{q_{nl}^i\}$. This set of descriptor components uses a feedforward neural network to construct the site energy which is given as[39,49,50]

$$U_i = \sum_{\mu=1}^{N_{neu}} w_\mu x_\mu - b, \tag{1}$$

where w_μ is the matrix of connection weights from the hidden layer to the output layer and b is the bias vector to the input layer. N_{neu} represents the number of neurons present in the hidden layer, and μ is the index for the neurons in that layer. x_μ is an activation function for the hidden layer and is given by

$$x_\mu = \tanh\left(\sum_{\nu=1}^{N_{des}} w_{\mu\nu}^0 q_\nu^i - b_\nu^0\right). \tag{2}$$

The NEP utilizes a hyperbolic tangent function as an activation function. N_{des} is the number of components of the descriptor vector, which is the sum of radial and angular descriptor components,[49] $w_{\mu\nu}^0$ is the matrix of connection weights from the input layer to the hidden layer, and b_ν^0 is the bias vector to the hidden layer. The descriptor component q_ν^i in Eq. (2), which are rotationally and translationally invariant, includes the radial and angular components of the NEP model. Equation (3) is the radial part of the descriptor, while Eqs. (5) and (6) are the angular description for the NEP model. Here, the radial descriptor is given using the expression below:

$$q_n^i = \sum_{j\neq i} g_n(r_{ij});\ 0 \le n \le n_{max}^R. \tag{3}$$

In Eq. (3), the radial function $g_n(r_{ij})$ is defined as a linear combination of $N_{bas}^R + 1$ basis functions $f_k(r_{ij})_{k=0}^{N_{bas}^R}$, with

$$f_k(r_{ij}) = \frac{1}{2}\left[T_k\left(2r_{ij}/r_c^R - 1)^2 - 1\right) + 1\right] f_c(r_{ij}), \tag{4}$$

where r_c^R is the truncation distance of the radial descriptor component and the coefficients c_{nk}^{ij} are adjusted during training.

The neighbor atoms i within the radial cutoff distance are considered during summation. Angular descriptor mainly consists of the three and higher body terms $q_{nl}^i(0 \le n \le n_{max}^A,\ 1 \le l \le l_{max}^{3b})$, and the four body term $q_{nl_1l_2l_3}^i(0 \le n \le n_{max}^A,\ 1 \le l_1 = l_2 = l_3 \le l_{max}^{4b})$ defined, respectively, as[39,49,50]

$$q_{nl}^i = \sum_{m=-l}^{l} (-1)^m A_{nlm}^i A_{nl(-m)}^i, \tag{5}$$

$$q_{nl_1l_2l_3}^i = \sum_{m_1=-l_1}^{l_1} \sum_{m_2=-l_2}^{l_2} \sum_{m_3=-l_3}^{l_3} \begin{pmatrix} l_1 l_2 l_3 \\ m_1 m_2 m_3 \end{pmatrix} A_{nl_1m_1}^i A_{nl_2m_2}^i A_{nl_3m_3}^i. \tag{6}$$

Here,

$$A_{nlm}^i = \sum_{j\neq i} g_n(r_{ij}) Y_{lm}(\theta_{ij}, \phi_{ij}), \tag{7}$$

where $r_{ij} = |r_j - r_i|$ is the coordinate difference vector of atoms i and j and Y_{lm}, θ_{ij}, ϕ_{ij} are the spherical harmonics, polar and azimuthal angle of r_{ij}, respectively.

In the training of the NEP model, the loss function is optimized with a separable natural evolution strategy (SNES),[51] and the total loss function L is defined as the sum of the weights of multiple loss functions,

$$L = \lambda_1 L_1 + \lambda_2 L_2 + \lambda_e \Delta U + \lambda_f \Delta F + \lambda_v \Delta W. \tag{8}$$

Here, ΔU, ΔF, and ΔW are root mean square errors (RMSEs) between predictions and reference values of energy, force, and virial, respectively. L_1 and L_2 are proportional to the first norm and second norm of the parameters in the model and λ_1, λ_2, λ_e, λ_f, and λ_v are the weighting terms.

B. Training NEP model

The Vienna *Ab initio* Simulation (VASP) density functional theory (DFT) package was employed to perform total energy and force calculations. A plane wave basis set with projector augmented wave (PAW) to characterize the ion–electron interaction was used along with the generalized gradient (GGA) approximation for the exchange-correlation potential parametrized by the Perdew–Burke–Ernzerhof (PBE) functional.[52,53] The energy cutoff was set to 520 eV, and convergence of electron self-consistent calculation was set to 10^{-6} eV. The Brillouin zone was sampled with a Γ-centered $2 \times 4 \times 4$ k-point mesh. The total dataset for the training of J14 consists of a total of 8888 structures. The structures are sampled using a supercell of $2 \times 1 \times 1$, thus consisting of 16 atoms: 8 of the atoms are cations Mg, Ni, Co, Zn, and Cu and 8 are oxygen atoms. Each eight cation with five atom selection for $2 \times 1 \times 1$ supercell results in a total of 35 such compositions. All of the 35 compositions of Mg–Co–Ni–Cu–Zn–O are sampled using special quasi-random structures (SQSs)[54] with displacements sampled from a canonical ensemble corresponding to temperatures of 800, 1375, and 1675 K, respectively. At least thirty static structures with random displacements are obtained for each of the temperatures 800, 1375, and 1675 K. Two different configurations using special quasi-random structures for each of the 35 compositions were generated. A total of 7758 five component structures ($2 \times 1 \times 1$ supercell) were generated; 6805 structures for the training and 953 structures for the testing. Furthermore, an additional 330 large supercells of size $4 \times 2 \times 2$ are included in the dataset, which consists of 297 structures in the training and remaining in the testing dataset. These supercells contain 13 and 12 metal atoms each, for a total of 64 metallic elements. The large supercells are deformed by a strain between -12% and $+12\%$ along xx, yy, zz, yz, xz, and xy, respectively. The DFT calculations are performed with Γ-centered $1 \times 2 \times 2$ k-point mesh. In addition to this, we also sampled five binary rock salt structures i.e., MgO, ZnO, NiO, CuO, and CoO. We sampled 100 displacement structures of each of these binary oxides. From a total of 500 structures, 450 structures were used for the training, and 50 structures were used for testing. Similarly, we also sampled all 10 possible ternary oxide combinations with metallic element concentrations of 25%–75%, 50%–50%, and 75%–25%. For each of these compositions, 10 snapshots were sampled from the canonical ensemble, producing a total of 300 snapshots of ternary compounds. 270 were used for the training and the remaining 30 for the testing. All the binary and ternary structures were from a $2 \times 2 \times 2$ supercell (64 atom supercell). The canonical ensemble sampling was similarly done at 800, 1375, and 1675 K. The same energy cutoff and convergence of self-consistent calculation as that of five component structures was used to perform DFT calculations. The Brillouin zone was sampled with a Γ-centered $4 \times 4 \times 4$ k-point mesh for these structures.

The dataset contains a wide range of compositions with different configurational disorders and displacements, some of them including binary and ternary compounds. The final training and testing datasets contain 7822 and 1066 structures, respectively.

C. Hyperparameters for NEP training

The training of the NEP model is done up to 2×10^6 generations, which is sufficient for six-component elements (iterations) to predict the energy, force, and virial from the DFT dataset. Here, in Fig. 1(a), we see that even though there are few gains, i.e., 0.27 meV/atom in the energy loss between 100 000 and 2.2×10^6 generations, there is a substantial gain in force from 0.1077 to 0.9503, which is about 12 meV/Å. It is worth noting that the computational cost of going from 100 000 to 2.2×10^6 generations, however, is 15 times larger. The total training time up to 100 000 generations took 16 GPU hours on a single NVIDIA RTX 4070 Ti. It is essential to train the model up to 2×10^6 generations to ensure that not only energy but loss in force and virial data are also sufficiently converged. These results are then compared with the corresponding DFT values to ensure the accuracy of calculation using parity plots from the NEP model and DFT calculation as can be seen in Fig. 1. In addition, the accuracy and convergence of the training are evaluated by RMSE values of energy, force, and virial; L_1 regularization; and L_2 regularization. The settings of the NEP training hyperparameters[49] are shown in Table I

The number of neurons is set to 60, and the weights of energy, force, and virial are $\lambda_e = 1.0$, $\lambda_f = 1.0$, and $\lambda_v = 1.0$, respectively. The number of structures per batch in the training is set to 400, and more than 2×10^6 generations are performed with 200 analytic solutions per generation. Based on the values of hyperparameters from Table I, the single hidden neuron layer network architecture of $61 - 60 - 1$ is obtained. Here, the first value in neural network architecture is due to the contribution from 11 radial descriptor components based on n_{max} from Table I, and 50 angular descriptor components. Now, for the choice of a hidden layer, only 60 neurons are considered, and this is adequate because of the sophisticated descriptor used in NEP and also to note that other machine learning potentials, such as deep neural networks, usually require a large number of neurons in the hidden layer due to its simple descriptor formulation.[37,39,55] The total number of parameters to be optimized is 1360913 609, and, therefore, the number of generations is set up to 2.25×10^6, which is sufficient for the given total number of parameters and types of atoms used. Using the values of the hyperparameters in Table I, the training is performed for J14, which is shown in Fig. 1(a). The loss functions in Eq. (7) are plotted, which shows that with the increasing generation, the root mean square error (RMSE) of the training dataset for energy, force, and viral converged to about 5.60 meV, 97.90 meV/Å, and 45.67 meV/atom, respectively. Similarly, the RMSE for the validation of the testing dataset converged to about 5.77 meV/atom, 98.51 meV/Å, and 45.82 meV/atom, respectively, which is in agreement with the values for high entropy alloys and similar ceramics machine learning potentials.[56,57] Considering the loss function L_1, L_2 in Eq. (8), first increases, which shows that the magnitude of the weight and bias parameters in the neural network first increases and then decreases, showing the effectiveness of regularization, which means setting λ_1 and λ_2 to a smaller value usually improve the fitting and not lead to the overfitting. The parity plots for per atom energy, force, and virial in Figs. 1(b), 1(c), and 1(d) show that the predicted energy, force, and virial using the NEP model is in agreement with the DFT energy, force, and virial, respectively.

III. RESULTS AND DISCUSSIONS

The smoothness of NEP potential is demonstrated by plotting the potential energy of the J14 single-phase crystal. The $5 \times 5 \times 5$ supercell of the ideal rock salt structure of J14 is generated with

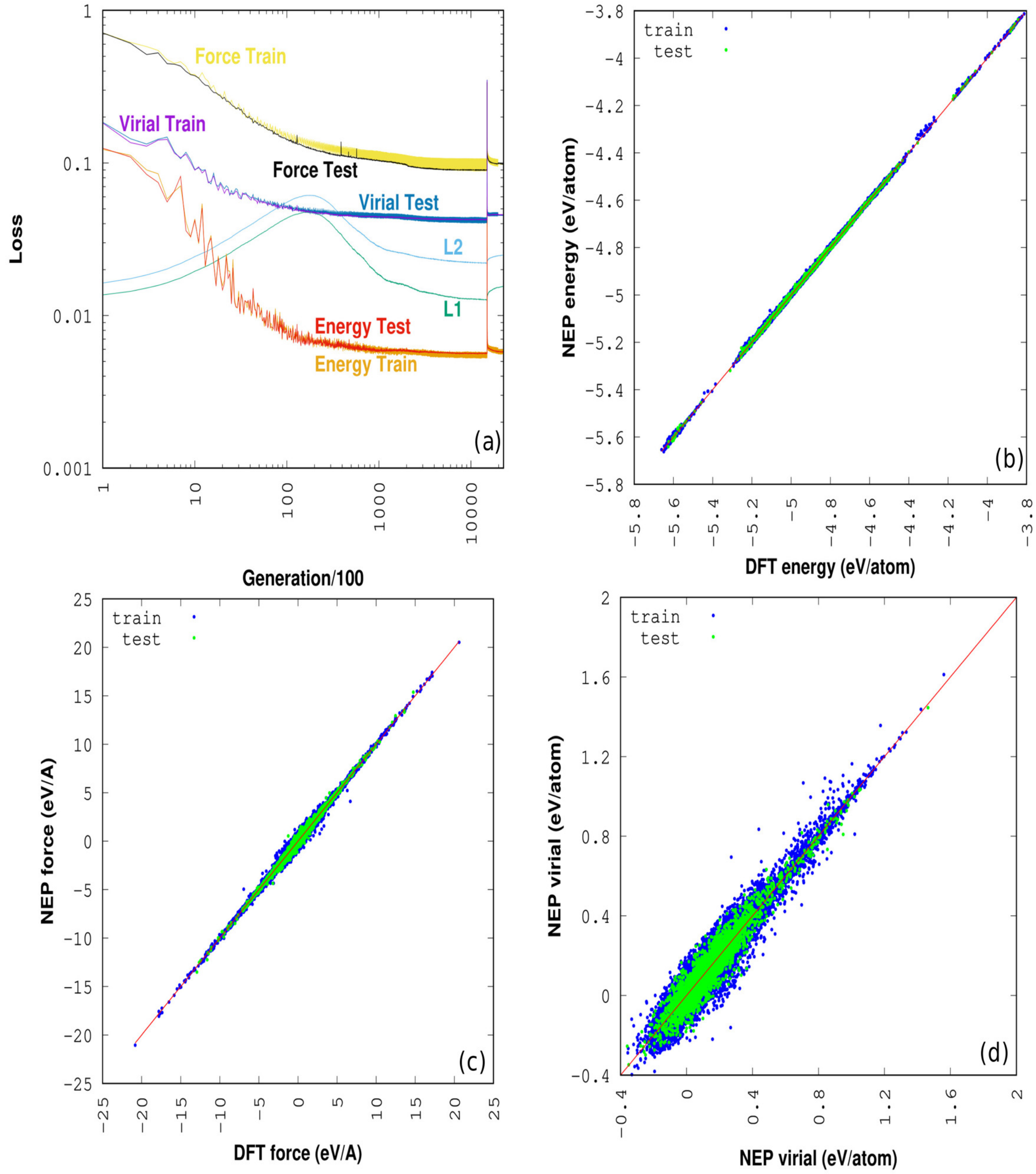

FIG. 1. (a) NEP training for J14 showing the loss functions with respect to a number of generations. L1 and L2 are the first and second norms in Eq. (7). (b) A parity plot for NEP training shows the predicted NEP energy and DFT energy in eV/atom for training and testing datasets. (c) Parity plot for NEP training showing predicted NEP force with the DFT force in eV/Å. (d) A parity plot for NEP training shows the predicted NEP virial with the DFT virial in eV/atom.

equal composition for five transition metals (TMs) using SQS, and the structure is relaxed (both volume and internal relaxation) using the calorine package as implemented in GPUMD.[49,54] After relaxing the $5 \times 5 \times 5$ supercell (lattice constant = 21.2573 Å), every of four sites of the elements, Mg, Ni, Co, Zn, Cu, and O, are randomly selected and are displaced up to ± 1 Å, respectively, from their relaxed position along the *x*-direction. One hundred displacement values are created for each of these randomly selected four sites for each element, and the total potential energy is calculated using NEP potential using GPUMD. Due to the difference in the ratio of

TABLE I. Hyperparameter setting of the NEP for J14.

Keywords	Parameters
Version	3
Type	6 Mg Ni Co Zn Cu O
Cutoff	8.4 3.9
n_{max}	10 9
Basis-size	12 12
l_{max}	4 2 0
Neuron	60
λ_1	0.05
λ_2	0.05
λ_e	1
λ_f	1
λ_v	1
Batch	400
Population	200
Generation	2 258 600

cations, static lattice distortion is expected to be observed compared to the ideal rock salt crystal structure. Figure 2 shows the potential energy of each transition metal (Co, Ni, Cu, Zn, Mg) and oxygen. The energy of the relaxed structure of the same $5 \times 5 \times 5$ supercell is subtracted from the energy of the displaced structure. Therefore, zero displacement distance is the ideal rock salt position. In Fig. 2, we see that the potential energy deviates from the harmonic behavior in the range of 0.3 Å. Some specific sites for cobalt, such as the 33rd site, have larger distortion than the ideal rock salt structure. This means that large local site disorder is more prominent around that atom. The average static lattice distortion for the cobalt obtained is of the order of 0.1 Å. Furthermore, we can see that a larger variation of the energy landscape is observed for Zn, Ni, and then Cu atoms as compared to Co and Mg atoms for the same atomic displacement.

Similarly, in Fig. 2, the potential energy of the copper atom shows that the minimum falls within ± 0.11 Å. Only four atomic sites are considered from the 100 copper sites in the $5 \times 5 \times 5$ supercell. The minimum for some sites may fall below this range.

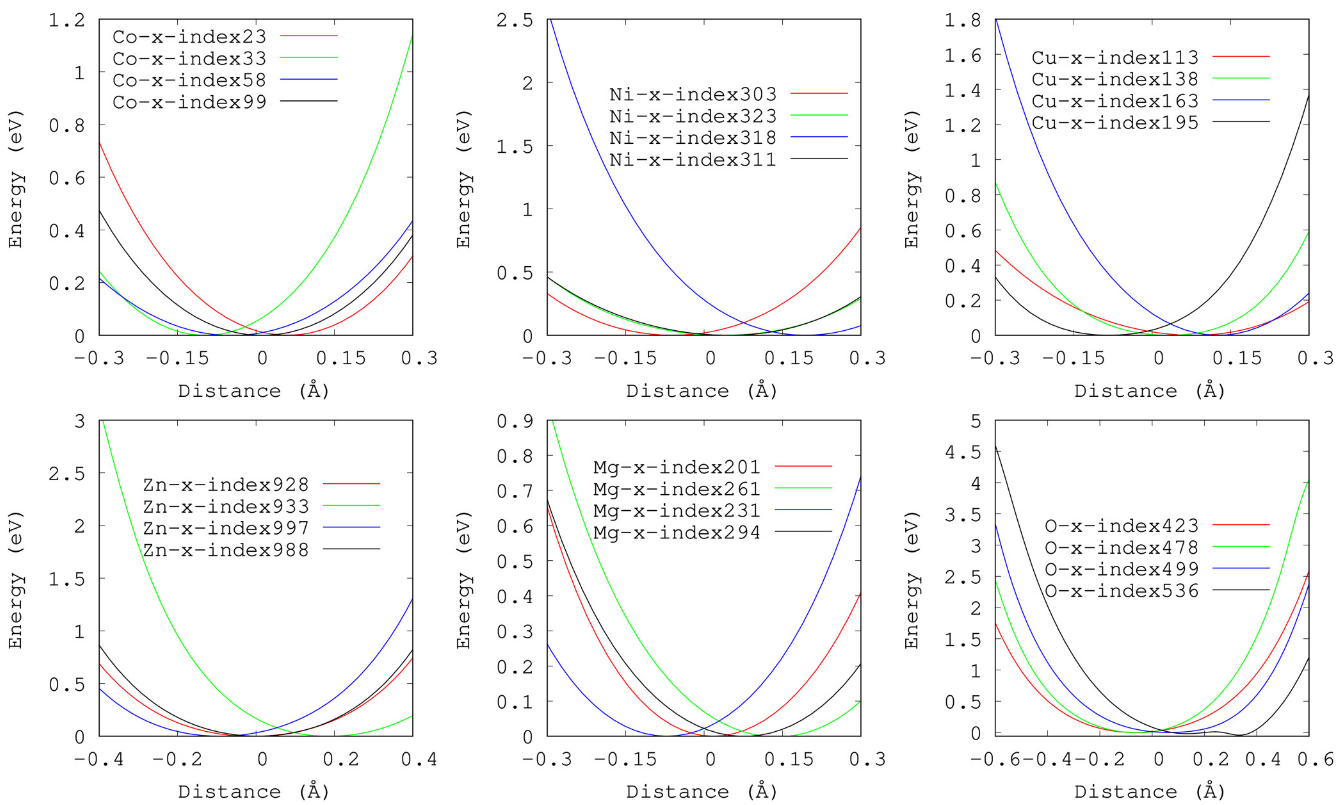

FIG. 2. Potential energy surface of each transition metal (Co, Ni, Cu, Zn, and Mg) and oxygen in J14. Here, we plot the energy surface by displacing each transition metal and oxygen atom along the *x*-direction in the relaxed structure by ± 1 Å. Four atoms are picked randomly, and the energy surface is plotted by displacing each atom along the *x*-direction. The *y*-axis represents the energy difference between the relaxed $5 \times 5 \times 5$ supercell structure of J14 and the one with the displaced structure so that the zero in the *y*-axis is the energy of the relaxed structure. Similarly, the *x*-axis represents the distortion with respect to the ideal rock salt position.

Note that the energy minima for the copper site are more spread around the ideal rock salt position than the cobalt site in Fig. 2, where the minima of cobalt are more concentrated around the ideal rock salt position. In Fig. 2, all the transition metals result in a single energy minimum. However, for the oxygen atom, it is interesting to note that the oxygen atom at the 536th site shows double well potential energy at 0.18 and 0.34 Å. It is difficult to explain which nearest neighbor transition metal environments have resulted in oxygen atoms having double energy minima; due to the double energy minima, the oxygen atom hops between sites that are attributed to order-disorder transition, which is beyond the scope of this study.[58]

It is essential to understand the role of thermal fluctuations on the lattice distortion in J14, besides distortion due to site disorder. Molecular dynamics simulations at temperatures of 10, 100,. . ., and 800 K are performed to study the lattice distortion at high temperatures further. For each temperature, constant volume, and temperature (NVT) runs were performed to obtain the distortion due to thermal fluctuations. This is followed by constant pressure and temperature (NPT) simulation to ensure the pressure is zero prior to performing NVT simulation. For the NPT simulation, the Berendsen thermostat, as implemented in GPUMD, is used, and only the hydrostatic pressure is allowed. The temperature coupling parameter is set to 100, and the pressure coupling constant is set to 2000. The time step of 1 fs is used for the NPT simulation for up to 100 ps (100 000 MD steps) to allow the $5 \times 5 \times 5$ supercell to reach thermal equilibrium. Similarly, for NVT simulation, the same time step and temperature controlling parameter are used for NPT simulation. The total simulation time of 300 ps (300 000 MD steps) is used, and the structure is recorded for every 100 time steps. Thus, 3000 structures are obtained for each of the given temperatures. The average distortion for each of the TMs and oxygen is evaluated using the expression below:[58]

$$\delta_\tau = \frac{1}{N_\tau . t_{max}} \times \sum_{i=1}^{N_\tau} \sum_{t=1}^{t_{max}} \sqrt{(x_{i,\tau}(t) - x^0_{i,\tau})^2 + (y_{i,\tau}(t) - y^0_{i,\tau})^2 + (z_{i,\tau}(t) - z^0_{i,\tau})^2}, \tag{9}$$

where τ refers to the atom type (Mg, Ni, Co, Zn, Cu, O), N_τ refers to the total number of atoms sites present for τ, t_{max} refers to the maximum timesteps used for the simulation, which in this case is (300 000 MD steps), $x^0_{i,\tau}$, $y^0_{i,\tau}$, $z^0_{i,\tau}$ refers to the undistorted rock salt lattice site i for atom type τ, and $x_{i,\tau}(t)$, $y_{i,\tau}(t)$, $z_{i,\tau}(t)$ is the lattice position at any given time t, respectively.

In addition to the energy surface of J14, the energy volume curve of two binary oxides, MgO and NiO, have been computed to validate the NEP potential for five component entropy stabilized oxide. Even though no training and test data were used for pure MgO and NiO, the NEP potential is able to fairly represent the energy volume curve for binary MgO and NiO. The energy volume curve is calculated up to 4% of the relaxed lattice constant for the ideal rock salt structure of MgO and NiO. As can be seen in Fig. 3(a), a good agreement is obtained for the energy volume curve for MgO. In Fig. 3(b), there is a shift in energy for NiO as compared to DFT energy. For each of the binary oxides, the energy volume curve is fitted with a third-order Birch–Murnaghan equation of state (EOS) using the vaspkit package.[59,60] The obtained bulk modulus of MgO using DFT and NEP potential based on this work are 149.53 and 161.07 GPa, respectively, which is in agreement with several other works.[61,62] Similarly, for the NiO, the bulk modulus

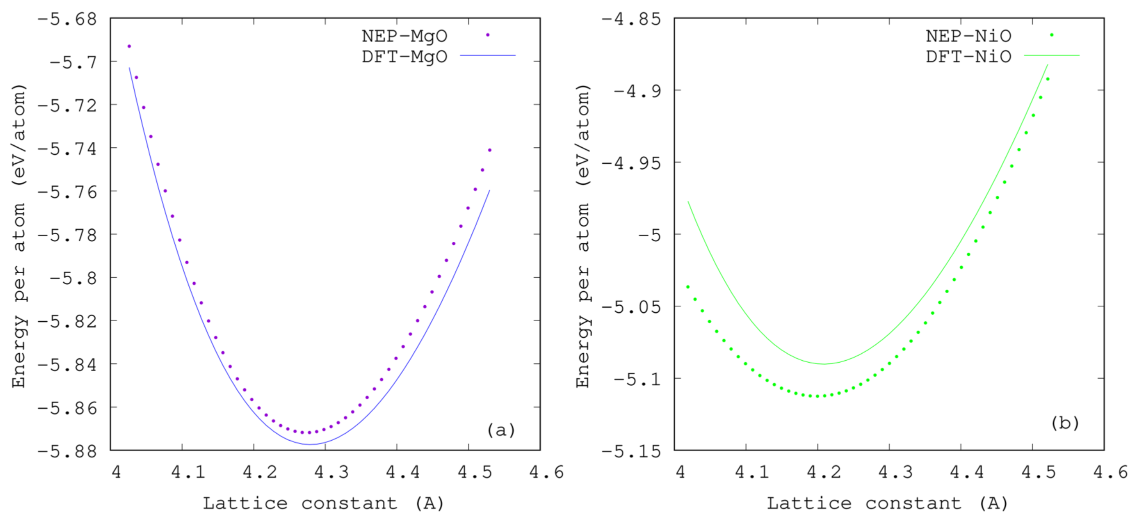

FIG. 3. (a) Energy volume curve for binary oxide of MgO using density functional theory and NEP potential. (b) Energy volume curve for binary oxide of NiO using density functional theory and NEP potential. The *y*-axis for figures (a) and (b) represent energy in eV/atom, and the *x*-axis is lattice constant in Å. Each of the energy volume curves for two binary oxides, MgO and NiO, are fitted using the third-order Birch–Murnaghan equation of state.

using DFT and NEP potential computed are 182.51 and 172.47 GPa, respectively, which is in agreement with the reported calculation.[63]

1. Static and dynamic distortions

To better understand the correlation between the distortion of oxygen atoms and the transition metals, the contribution from thermal fluctuation for each transition metal and oxygen has to be decoupled because of the complex cationic interactions present due to local site disorder. In Fig. 4, the average site distortion with respect to temperature has been plotted for the TMs and the oxygen atom. For simplicity, the magnitude of distortion from the ideal rock salt structure is presented at temperatures of 10, 300, and 800 K, and the average site distortion for all TMs and oxygen is given in Table II. For the distortion of Cu, Mg, Ni, and Zn, the temperature values have been shifted by 10, 20, 30, and 40 K so that they do not overlap with the distortion plot of the Co atom. While the distortion at 10 K for all TMs is almost similar, i.e., about 0.1 Å, the maximum distortion for each TM varies largely from 0.2 to 0.3 Å at 10 K in the Fig. 4. This suggests that the distortion for each TM is highly environment-dependent since, at a low temperature of 10 K, almost minimal thermal fluctuation is present. Considering the ratio between minimum and maximum site distortion, the ratio for TMs is about 10 and about 17 for oxygen at 10 K, which suggests that there is strong static lattice distortion present in entropy-stabilized oxide. The average distortion for each TM and oxygen increases up to 300 K and begins to saturate afterward. As can be seen in Fig. 4, at low temperatures, there is almost no thermal fluctuation present, so both the minimum and maximum site distortions increase. At higher temperatures, since TMs do not distort much further, the maximum distortion tends to saturate. However, the minimum site distortion keeps on increasing. One of the possible reasons why the maximum distortion further decreases as temperature increases is to maintain the

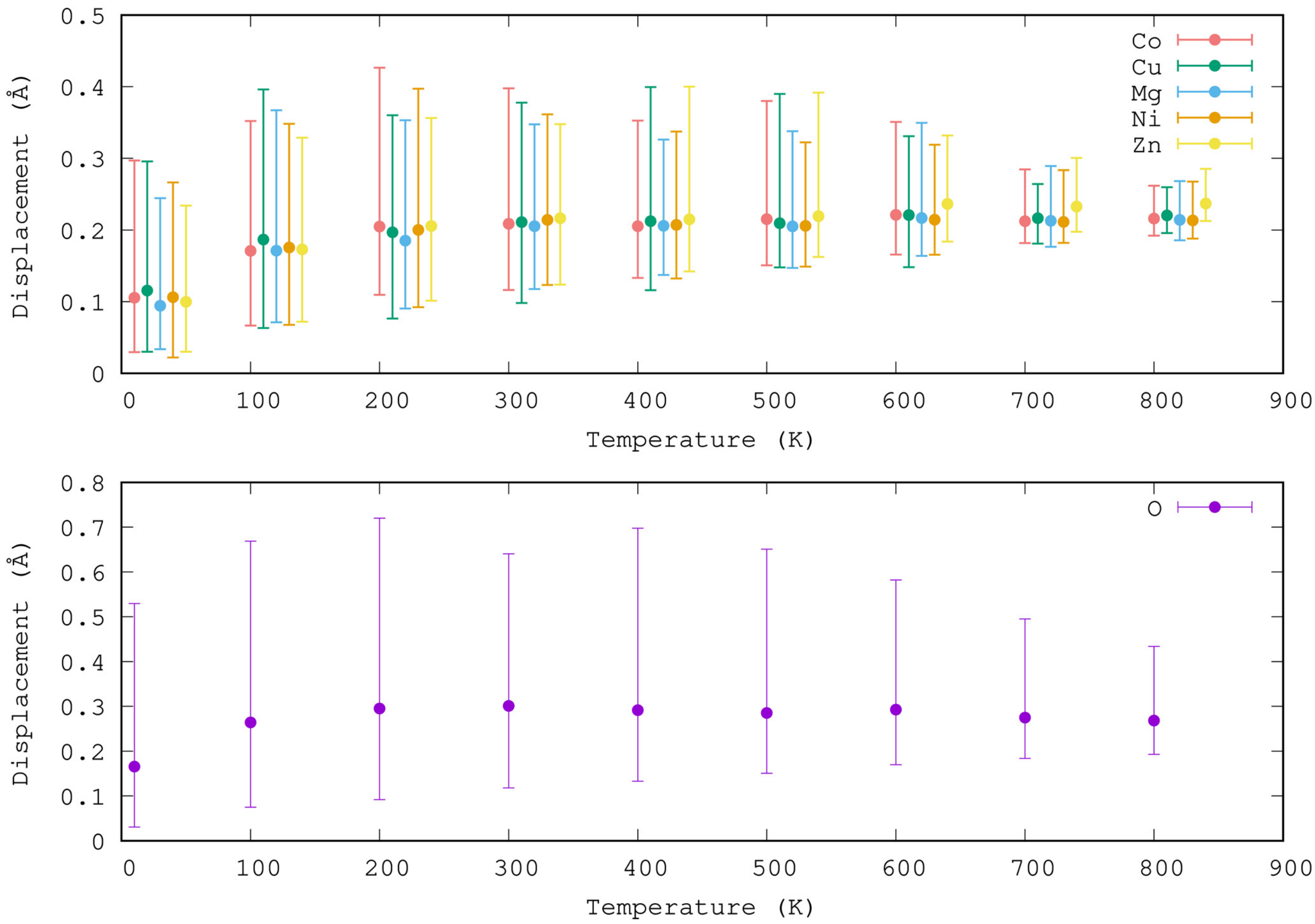

FIG. 4. Temperature dependent lattice distortion of transition metals (TMs) and oxygen from 10 to 800 K. Here, the circular dot represents the average distortion of each transition metal on the upper figure and oxygen on the lower figure. For clarity, the Cu, Mg, Ni, and Zn temperatures have been shifted to avoid overlap with Co data. The error bars represent the smallest and largest site distortion of TMs and oxygen at that temperature.

TABLE II. Site dependent statistical averages of distortion (in Å) for each element based on 300 ps molecular dynamics (MD) snapshots.

	Temperature (K)	δ_{Mg}	δ_{Ni}	δ_{Co}	δ_{Zn}	δ_{Cu}	δ_{O}
Min. site distortion	10	0.034	0.022	0.029	0.030	0.030	0.031
	300	0.117	0.123	0.116	0.123	0.098	0.118
	800	0.186	0.188	0.192	0.212	0.196	0.193
Avg. site distortion	10	0.094	0.106	0.105	0.099	0.116	0.165
	300	0.205	0.214	0.208	0.216	0.211	0.301
	800	0.214	0.213	0.216	0.237	0.220	0.268
Max. site distortion	10	0.244	0.266	0.297	0.234	0.296	0.529
	300	0.347	0.361	0.398	0.347	0.378	0.641
	800	0.268	0.268	0.262	0.285	0.259	0.434

octahedral coordination environment and bond length. Because of this competing effect of sites with minimum distortion increasing with temperature and the sites with maximum distortion decreasing with temperature, we can see that the average distortion for TMs saturates at about 0.2 Å at higher temperatures. A similar trend as that of TMs is observed for oxygen atoms. The difference is that the average distortion for oxygen atom saturates to a larger value of 0.3 Å. While the site with minimum distortion for oxygen atoms is similar to that of TMs, the site with maximum distortion in oxygen atoms has a larger distortion by a factor of 1.5 compared to TMs.

To better understand the size of distortion for each individual transition metal, a histogram containing molecular dynamics snapshots of sites with minimum and maximum distortion is plotted in Fig. 5. The top row (from left to right) represents the histogram of the time evolution of the site with minimum distortion at temperatures of 10, 300, and 800 K, respectively. Here, at a low temperature of 10 K, the histogram with Mg and Zn are shifted to the right with their mean toward a large distortion value, and the one with the lowest distortion (the mean of Gaussian distribution toward the small displacement value) is Ni. At 10 K, the Cu and Co distortions fall in between Zn and Ni. Since the binary zinc oxide and copper oxide do not form a cubic rock salt structure, it is likely that the zinc and copper atoms should distort more. The minimum distortion of the Mg atom is similar to that of the Zn atom. Also, considering the maximum distortion of Mg and Zn atoms (bottom row, first figure in Fig. 5), it is clear that these atoms tend to maintain lower variation on distortions within the entropy-stabilized oxide. For Cu and Co atoms, the maximum distortions are the greatest, and the minimum distortions are the smallest at 10 K, which suggests that large variations in distortion are present for these atoms at low temperatures. It is likely that the contribution of the maximum distortion for Cu and Co atoms is mainly to accommodate the Zn atom to the octahedral coordination since the binary zinc oxide does not form a rock salt structure even at low temperatures.

Now, for the oxygen atom, it is clearly seen in Fig. 5 that it is dominated by a high degree of distortion relative to TMs, in agreement with experimental results of Rost *et al.*[64] Since the mean of the histogram for minimum distortion for the oxygen atom is similar to that of TM at 10 K, the stark difference in maximum distortion peaks between TMs and the O atom might have resulted in the overall average distortion of the oxygen atom increasing. It is important to address the static distortion (distortion at low temperature) first because the thermal fluctuation only increases the width of the Gaussian distribution, therefore shifting the distortion to a larger value as it is clear from Fig. 5 that the nature of minimum and maximum distortion is equivalent for the temperature 10, 300, and 800 K (left to right) but only the width of the gaussian distribution increases. It is also observed from the Fig. 5 that even though Cu has a maximum distortion value as the Gaussian distribution shifts toward the larger displacement (bottom row middle figure) at 300 K, it is interesting to note that Zn distorts more at 800 K rather than Cu atom or other TMs. It is possible that these specific distortion averages are obtained for the specific supercell decoration adopted in this study. But the fact the TM distortions are systematically lower than O remains universal. Indeed, the oxygen environment is more diverse than the TM ones, which all have oxygen as their nearest neighbor.

An alternative to studying distortion with respect to the average rock salt position is to consider the bond length or pair separation distances represented by the radial distribution function (RDF). The first nearest neighbor to a TM is oxygen, while the second nearest neighbor is another transition metal. As an illustration, in Fig. 6(a), the RDF for Mg–O and Zn–O pair is plotted as a function of interatomic distance for temperatures of 10, 300, and 800 K, respectively. Peaks are sharper at lower temperatures, indicating more order. As temperature increases, there is a decrease of the RDF peak from 10 to 800 K for the Mg–O, Zn–O, and O–O pair in Figs. 6(a) and 6(b), respectively. Due to increased lattice distortion of all TM including, Mg and Zn, and oxygen from 10 to 800 K in Fig. 4 the Mg–O and Zn–O bonds stretches or compress to accommodate for ideal rock salt position and simultaneously rattle to account for thermal fluctuation thereby increasing the width of the peak. It can be seen from Fig. 6(a) that the Zn–O bond length peak is broader as compared to the Mg–O bond length for the first nearest neighbor, implying that Zn–O has a wider variation in bond length even at 10 K. Also, it is true for the case of 800 K that the Zn–O bond length varies more compared to the Mg–O pair. The first nearest neighbor distance for Mg, O and Zn, O pair are 2.0425 and 2.0516 Å, while the pair varied by 0.334 and 0.347 Å, respectively. To understand the effect of temperature on the oxygen–oxygen pair, the RDF for the oxygen–oxygen pairs (second nearest neighbors) is plotted in the Fig. 6(b) where the bonds are slightly stretched as the temperature is increased.

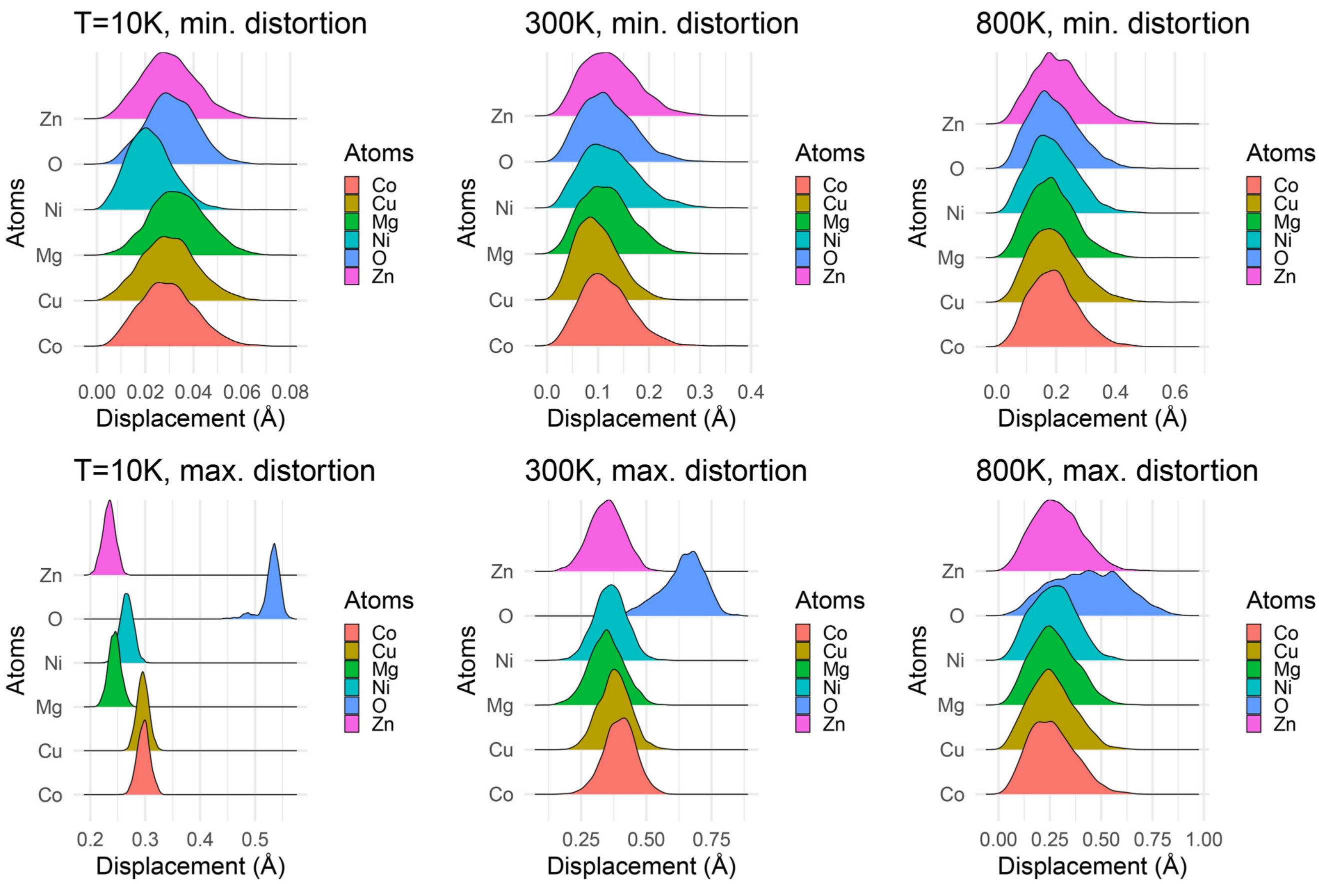

FIG. 5. Histograms for the molecular dynamics snapshots of the minimum and maximum displacement of each atom in the plot of displacement vs temperature (in the above Fig. 4). The first row represents the histogram of atoms of minimum displacement at 10, 30, and 800 K, respectively, and the latter represents the histogram of atoms of maximum displacement at 10, 300, and 800 K, respectively.

2. Mechanical properties

In addition to high-temperature lattice distortion, the high-temperature deformation of entropy-stabilized oxide of J14 has been studied using stress–strain response. The normal tensile strain is applied separately to the same $5 \times 5 \times 5$ supercell of J14 along the *x*, *y*, and *z* directions, respectively. For each of the strain directions, only the longitudinal dimension is allowed to change, and the lateral or transverse dimension is fixed. The strain is incremented by steps of 0.6452% up to the final strain value of 12%. A similar setup is repeated for each temperature of 10, 300, 600, and 900 K. At any given temperature, and prior to applying the strain on the supercell, it is first relaxed at zero pressure (NPT) by applying a Berendsen thermostat. A temperature coupling parameter value of 100 and a pressure coupling constant of 2000 are used in this case. The timestep used is 1 fs, and the supercell is allowed to relax for 100 ps (100 000 MD steps). Then a strain is applied, and the pressure is measured. For each data point, the volume of the supercell is fixed, and constant volume and temperature (NVT) is performed using the Nosé–Hoover chain method in GPUMD. The trajectory with stress information is recorded at each 100 MD steps, and the simulation is allowed to run for 50 ps (50 000 MD steps). This generated 500 structures and 500 sets of stress components upon which the average is taken as the stress response.

For the stress–strain response at each temperature, 10, 300, 600, and 900 K, linear polynomial fitting is performed up to the ultimate strength, which is the maximum value of the stress in Fig. 7 for all temperatures. It is interesting to note that, unlike in metals, the stress–strain curve for each temperature has a very small region of nonlinearity around which it would deform plastically before it breaks. Also, note that the breakpoint occurs at slightly different strain values for each direction due to different configuration disorders present in each direction. The effect of temperature on the breaking of entropy-stabilized oxide is directly related to the strain; i.e., at larger temperatures, smaller strains are required to break the sample. Also, in the inset of Fig. 7, the plot of the ultimate tensile strength can be seen. From the stress–strain curve, we pick the ultimate tensile strength for each temperature by

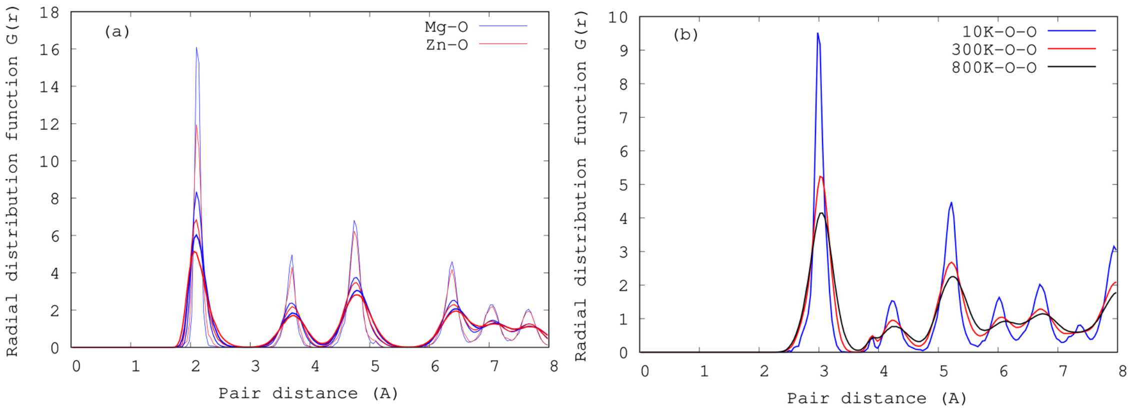

FIG. 6. (a) Radial distribution function (RDF) of Mg, O and Zn, O in J14. The decreasing peak of each colored line represents RDF at 10, 300, and 800 K, respectively0. (b) Temperature dependent RDF plot for oxygen pairs in J14.

taking the smallest of the three maximum values of stress along the x, y, and z directions. It is clear in the stress–strain curve in Fig. 7 that the upper bound to the yield strength almost overlaps with the ultimate tensile strength point. This is because there was not enough time for dislocation motion to be generated so that the yield stress is clearly visible. Each colored point in the inset represents the ultimate tensile strength at that particular temperature. The plot of ultimate tensile strength follows a linear decreasing trend, which is then fitted as a function of temperature given by the black solid line. The average is taken over the three cartesian

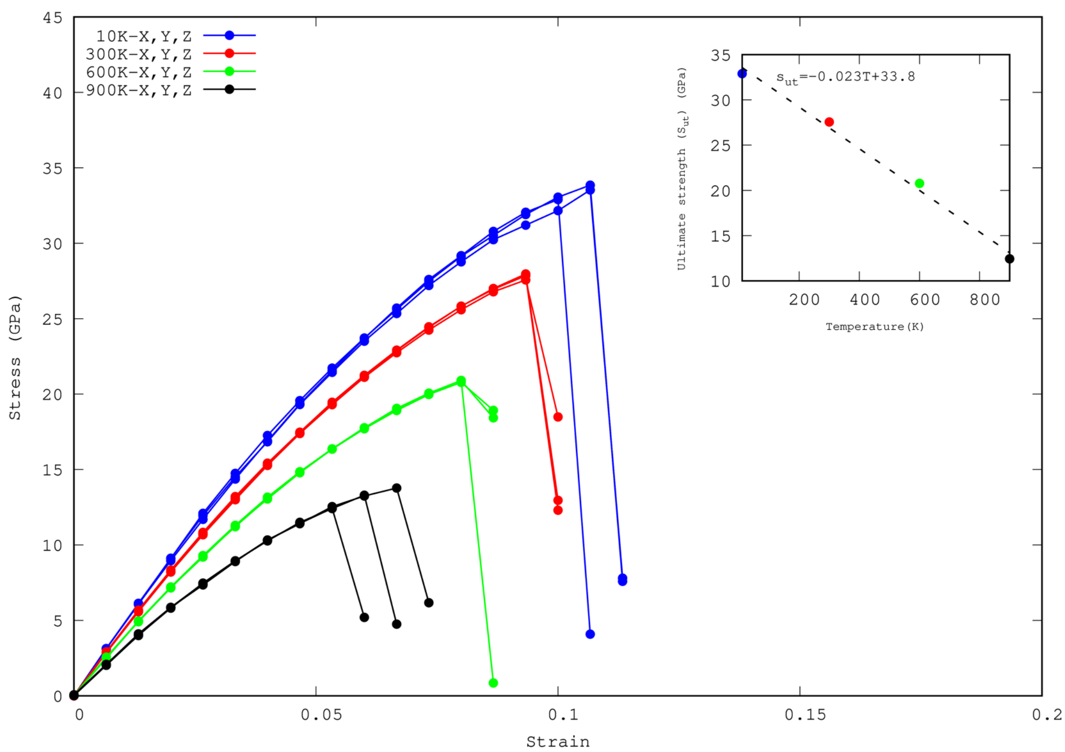

FIG. 7. Stress response of J14 at various temperatures. The 5 × 5 × 5 supercell is used in tension at varying strains up to 0.12 along x, y, and z directions, respectively. Different colored curves represent the stress–strain response at different temperatures and tension directions. The inset graph represents the ultimate tensile strength corresponding to that temperature. The dashed line in the inset plot is the linear fitting of ultimate tensile strength values.

TABLE III. Elastic constants, lattice constants, and modulus for various temperature.

Temp. (K)	*a* (Å)	C_{11}	C_{11} [65]	C_{12}	C_{12} [65]	*B* (GPa)
10	4.25	330.05 ± 3.74	303	113.26 ± 15.65	124	185.52
300	4.27	300.07 ± 1.44	…	104.27 ± 16.63	…	169.54
600	4.30	262.19 ± 0.76	…	92.69 ± 16.84	…	149.43
900	4.33	221.43 ± 0.62	…	79.56 ± 17.99	…	127.14

directions in order to calculate the elastic moduli, which are given below in the table. The standard deviation is shown for each of the averages in Table III. The elastic constant C_{11} or Young's modulus at 10 K for the *x*, *y*, and *z* directions obtained by linear fitting are 331.29, 324.98, and 333.88 GPa, respectively. The sum of square error or the residuals for each of the linear fits are 1.49, 1.55, and 1.42. A similar calculation for 300 K shows that the elastic constant C_{11} using three independent *x*, *y*, and *z* strain directions are 300.80, 298.07, and 301.35 GPa with the fitting residuals 1.27, 1.29, and 1.25, respectively. For temperature 600 K, the obtained values of C_{11} are 262.52, 261.14, and 262.92 GPa for the *x*, *y*, and *z* directions with the residuals 1.004, 0.996, and 0.990, respectively. Finally, for 900 K, the values of C_{11} are 221.077, 220.899, and 208.859 GPa, respectively, for the *x*, *y*, and *z* directions, respectively. The residuals of the fit for the temperature of 900 K are 0.63, 0.634, and 0.81. The previous DFT study of the elastic constants and bulk modulus of J14 is also shown in Table III.[65] In this context, this value is the ground state elastic constant of non-magnetic J14 and not at 10 K. A similar linear fitting technique is used for obtaining the elastic constant C_{12}, and the average value is given in Table III with the standard deviation. The bulk modulus for J14 is obtained assuming cubic crystals. The lattice constant as a function of temperature is also given in Table III. The estimated value of the lattice constant at 300 K using NEP potential is 4.27 Å. An experimental study by Rost *et al.* reported the lattice constant of J14 to be 4.237 Å at room temperature, which is in agreement with the current study.[64]

The temperature dependence of elastic constants is usually explained by using the empirical relation $C_{ij}(T) = C_{ij}^0 - s/(e^{t/T} - 1)$,[66] where C_{ij}^0 is the elastic constant at zero temperature and *s*, *t* are fitting parameters. Based on our simulation of the temperature-dependent stress–strain curves, the elastic constants C_{11} and C_{12} are plotted as a function of temperature in Fig. 8. Since it is usually the case that the elastic constants for a given crystal go down as a function of temperature and also based on the empirical relation, we fit the elastic constants C_{11} and C_{12} using a linear fit (dashed colored line in Fig. 8). If the parameter *t* is small, the empirical relation can be expressed using a linear relation which is the dominant high-temperature limit of the above equation. The fitted dashed blue line represents the temperature-dependent elastic constant for C_{11}, and the red dashed line represents the elastic constant for C_{12} in Fig. 8. The ground state elastic constants for C_{11} and C_{12} obtained from the fit are 333.88 and 114.62 GPa, respectively.

3. Thermal conductivity of a binary $MgNiO_2$

The lattice thermal conductivity of $MgNiO_2$ is computed using the NEP potential as an example. Homogenous nonequilibrium molecular dynamics (HNEMD), as implemented in GPUMD, have been used to evaluate lattice thermal conductivity.[67] The HNEMD method is constructed for many body potentials based on the linear response theory, and the lattice thermal conductivity is evaluated using the expression[67]

$$\kappa(t) = \frac{\langle J_q(t) \rangle_{ne}}{TVF_e}, \tag{10}$$

where $\kappa(t)$ is the running thermal conductivity, $J_q(t)$ is the heat current, T is the temperature, V is the volume of the simulation cell, and F_e is the driving force parameter in the HNEMD formalism, respectively. For computing the lattice thermal conductivity of a solid solution of $MgNiO_2$, a supercell size of $10 \times 10 \times 10$ having 8000 atoms is used with an equal composition of Mg and Ni atoms. For each of the temperatures, the heat current is measured using an NVT ensemble during 10 ns. Five independent runs were made, each with 10 ns production time for the ensemble average of heat current. Regarding the choice of F_e, it is essentially varied until κ does not depend on it. Its optimal value changes with T and is shown in Fig. 9(a).

The figure shows an almost linear relation between the driving force value and temperature. Previous studies have shown that the driving force parameter value should be smaller than the inverse of

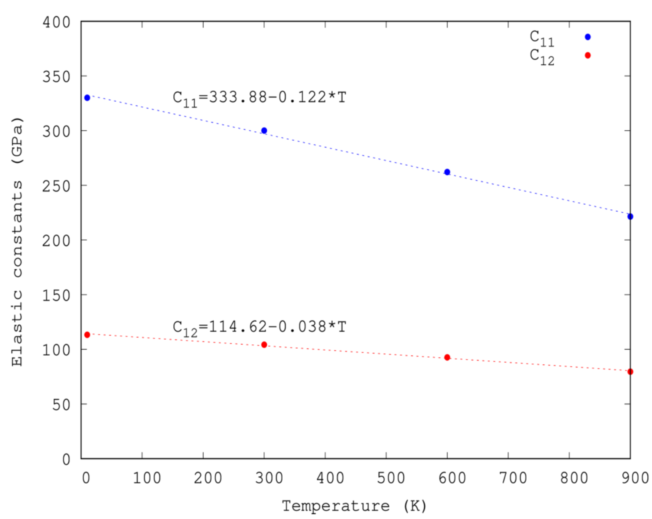

FIG. 8. Variation of elastic constants with temperature. The dashed lines represent the fitting of elastic constants, C_{11} and C_{12} using the linear expression $C_0 + m^*T$. The parameter C_0 is the elastic constant at the ground state, and *m* is the slope.

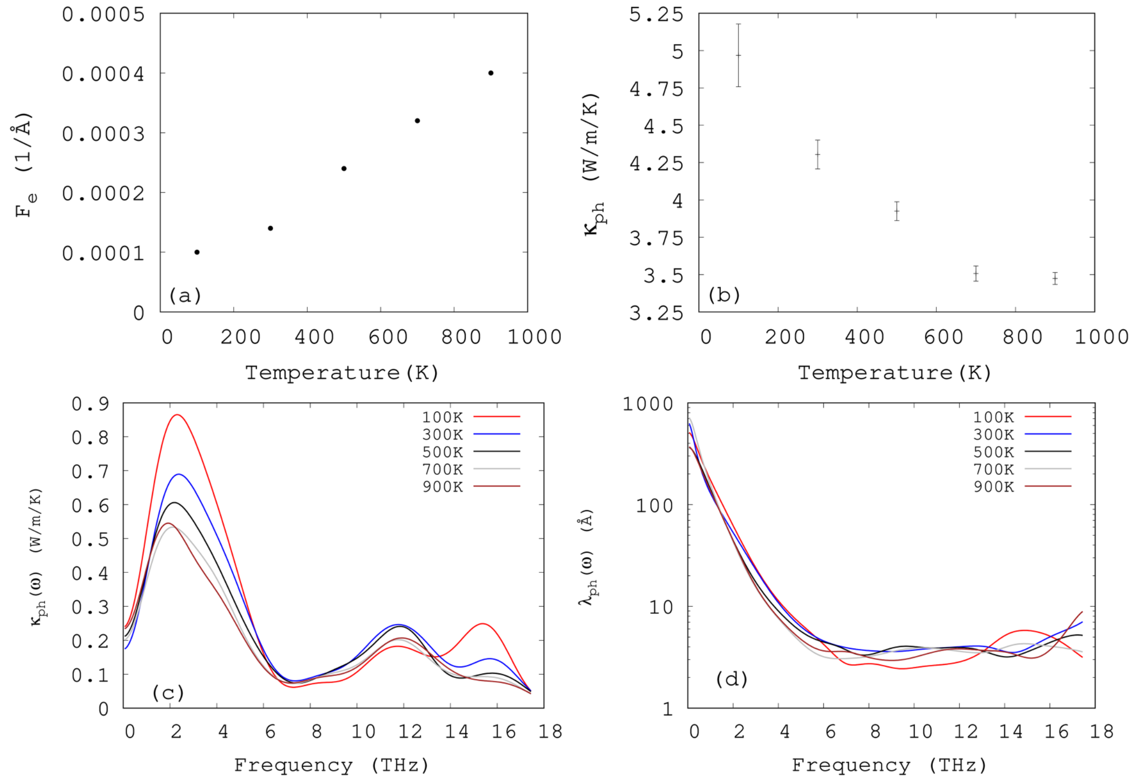

FIG. 9. (a) Driving force (F_e) as a function of temperature. (b) Lattice thermal conductivity of $MgNiO_2$ using NEP potential as a function of temperature. (c) Corresponding spectral thermal conductivity as a function of frequency. (d) Frequency dependent phonon mean free path of $MgNiO_2$ at different temperatures. Note the *y*-axis representing the phonon mean free paths is in the logarithmic scale.

the mean free path of the system under study.[68,69] The lattice thermal conductivity for different temperatures is displayed in Fig. 9(a). The room temperature lattice thermal conductivity of $MgNiO_2$ obtained using NEP potential for this study in Fig. 9(b) is 4.304 ± 0.097 W m^{-1} K^{-1}. Several studies have reported the lattice thermal conductivity of pure MgO to be around 10–15 times this value.[70–72] Such a large reduction in lattice thermal conductivity in a solid solution is common and is due to strong scattering of heat carriers by site disorder and distortions. Recent studies have demonstrated that the reported lattice thermal conductivity for the J14 family ranges from 0.5–11 W m^{-1} K^{-1}.[26] To put the lattice thermal conductivity of $MgNiO_2$ into perspective, the reported value of thermal conductivity by Braun *et al.* for a two-cation oxide of $Cu_{0.2}Ni_{0.8}O$ and $Zn_{0.4}Mg_{0.6}O$ at room temperature, are 6.56 and 11.97 W m^{-1} K^{-1}, respectively. Furthermore, a thermoelectric properties study by Ruttanapun *et al.* has reported the thermal conductivity of equi-composition ternary oxide of $CuAlO_2$ to be 3.5 W m^{-1} K^{-1}.[73] Unlike J14, where the thermal conductivity trend behaves similarly to amorphous material, the lattice thermal conductivity of $MgNiO_2$ in Fig. 9(b) shows a decrease with temperature from 4.3 at 300 K to 3.5 W m^{-1} K^{-1} at 900 K. This shows that the extent of order is prevalent in the ternary system of $MgNiO_2$ up to a temperature of 700 K. This decrease is due to stronger scattering of low-frequency phonons as T is increased. To support this hypothesis, we show the spectral thermal conductivity of $MgNiO_2$ in Fig. 9(c). The latter is computed by Fourier transforming the nonequilibrium heat current $\langle J_q(t)\rangle$ as implemented in the GPUMD package.[49,74–76] We can also note in Fig. 9(d), the frequency-dependence of carrier mean free paths from 100 to 900 K. Depending on the temperature, they decrease from about 30 to 70 nm at very low frequencies to a few angstroms at and above 5 THz. This is the lowest possible limit as it is on the order of the lattice constant. While low-frequency heat carriers behave as propagons, above 5 THz, they are expected to be diffusions in mid-range, and locons at band edges. Strictly speaking, propagon MFPs are expected to decrease with *T*. However, due to statistical uncertainties in our simulations, this trend is partially visible.

IV. CONCLUSION

In this work, neuroevolution machine learning potential was constructed for the entropy-stabilized oxide J14 to study high-temperature lattice distortions and their effect on mechanical and thermal properties of this material. The training error and test errors for the energy, force, and virial of the developed NEP potential were 5.60 meV/atom, 97.90 meV/Å and 45.67 meV/atom and 5.77 meV/

atom, 98.51 meV/Å and 45.82 meV/atom, respectively. Molecular dynamics (MD) simulations have been performed to study the lattice distortion in J14. By using this potential, it was shown that oxygen atoms distort more than the transition metals elements from their ideal rock salt position, consistent with the experimental observation.[64] The characteristic feature of the distortion is that each cationic species (transition metals) and oxygen distortions first increase and then saturate at high temperatures. It was shown that this saturation occurs due to the competing effect between the site with minimum distortion that increases with temperature and the site with maximum distortion that decreases with temperature. Further examination based on the maximum site distortion snapshots reveals that at low temperatures where static distortion prevails, the Cu and Co atoms are more distorted considering the transition metals; however, at 800 K, the Zn atom also has higher distortion. In addition to the lattice distortion, the temperature-dependent stress–strain simulation results indicate that the mechanical properties like elastic constants, bulk modulus, and ultimate tensile strength decrease linearly with temperature. It was further shown with fitting that the elastic constant C_{11} decreases much faster as compared to C_{12} with respect to temperature. Elastic constants C_{11} and C_{12} were in agreement with the previous study.[65,77] The lattice thermal conductivity of $MgNiO_2$ is found to decrease from $4.25\,W\,m^{-1}\,K^{-1}$ at 300 K to $3.5\,W\,m^{-1}\,K^{-1}$ at 900 K due to strong scattering of low frequency modes at high temperatures.

The as-developed machine-learned neuroevolution potential is applicable to the study of entropy-stabilized oxide J14 for high-temperature lattice distortion and deformation. Neuroevolution potential has been successful in explaining the properties related to multicomponent and high entropy alloys, especially works related to thermal transport applications using equilibrium molecular dynamics and homogenous non-equilibrium molecular dynamics simulation.[47,78] Subsequent works based on the as-developed NEP potential for entropy-stabilized oxide J14 will be related to estimating the thermal transport properties, including but not limited to phonons, temperature, composition-dependent thermal conductivity, and spectral thermal conductivity of this oxide.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

B. Timalsina: Conceptualization (supporting); Data curation (lead); Formal analysis (equal); Funding acquisition (supporting); Investigation (lead); Methodology (supporting); Project administration (supporting); Resources (supporting); Software (lead); Supervision (supporting); Validation (equal); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **H. G. Nguyen:** Conceptualization (supporting); Data curation (supporting); Formal analysis (supporting); Funding acquisition (supporting); Investigation (supporting); Methodology (supporting); Project administration (supporting); Resources (supporting); Software (supporting); Supervision (supporting); Validation (equal); Visualization (equal); Writing – original draft (supporting); Writing – review & editing (supporting). **K. Esfarjani:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (lead); Investigation (equal); Methodology (lead); Project administration (lead); Resources (lead); Software (equal); Supervision (lead); Validation (equal); Visualization (equal); Writing – original draft (supporting); Writing – review & editing (supporting).

DATA AVAILABILITY

The processed data required to reproduce these findings are available from the corresponding author upon reasonable request.

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